

بسم الله الرحمن الرحيم

Average Poynting Vector:

$$S_{avg} = \frac{E^2}{240\pi}$$

Radiation Intensity:

$$I = \frac{E^2}{240\pi} \rightarrow E = \sqrt{240\pi \cdot I}$$

Radiation power:

$$\frac{P}{A} = \frac{E^2}{240\pi} \rightarrow E = \sqrt{240 \cdot \frac{P}{A}}$$

$$\mu_0 = 240\pi \text{ free space}$$

$$\mu_0 \mu_r = 240\pi \mu_r$$

↓
المواد

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⑤ An Airplane at 10 km receives a hemisphere radiation of $10 \mu\text{W}/\text{m}^2$

$$r = 10000 \text{ m}$$

$$I = 10 \times 10^{-6}$$

$$\textcircled{a} \quad E = \sqrt{240\pi \times I} = \sqrt{240\pi \times 10 \times 10^{-6}} = 0.087 \text{ V/m}$$

$$\textcircled{b} \quad B = \frac{E}{c} = \frac{0.087}{3 \times 10^8} = 0.029 \times 10^{-8} \text{ T}$$

$$\textcircled{c} \quad P = I \times A = 10 \times 10^{-6} \times 2\pi \times (10000)^2 = 6283.2 \text{ W}$$

$$\star 2\mu.c = 240\pi \text{ free space}$$

$$\star 2\mu.c \leftarrow \mu_r = 240\pi \times \mu_r$$

المغناطيسية

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⑥ 15 mW laser emits beam in circular radiation with diameter $\frac{2}{1000}$ m

$$a) E = \sqrt{240\pi \cdot \frac{P}{\pi r^2}} = \sqrt{\frac{240 \times 15 \times 10^{-3}}{(1 \times 10^{-3})^2}} = 1897.4 \text{ V/m}$$

$\times 2u.c = 240\pi$ free space

$\times 2u.c \times u_r = 240\pi \times u_r$

$\frac{1}{2} \times 10^{-3}$

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$$E = 20 \text{ V/m}$$

$$r = 10 \text{ m}$$

$$\textcircled{a} B = \frac{E}{c} = \frac{2}{3 \times 10^8} = 0.67 \times 10^{-8} \text{ T}$$

Ⓔ Radiation Intensity

$$I = \frac{E^2}{240\pi} = \frac{2}{240\pi} = ?$$

Q₆ medium $\rightarrow 40\%$
 $V_p = ?$

$$V_p = \frac{mv^2}{2q} = \frac{9.11 \times 10^{-31} \times (1.2 \times 10^8)^2}{2 \times 1.6 \times 10^{-19}}$$

$$V_p = 40995 \text{ V}$$

Q₇

$$E_y = 118 \sin((1.2 \times 10^3)x - \omega t)$$

$$B = \frac{E}{c} = \frac{118}{3 \times 10^8} = ? \text{ at } \textcircled{2}$$

$$\phi = \frac{c}{2\pi} \times k = \frac{3 \times 10^8}{2\pi} \times 1.2 \times 10^3$$

Question

$$B_z = E \sin \pi (2x/3 - \omega t)$$

Ⓐ $E = B \times c = 1 \times 3 \times 10^8 = 3 \times 10^8$

Ⓑ $\phi = \frac{c}{2\pi} \times k = \frac{3 \times 10^8}{2\pi} \times \frac{2\pi}{3} = 10^8$

$$F = q \vec{v} \times \vec{B}$$

$$S = \frac{1}{4\pi} \frac{E}{B}$$

Q₈ $I = 1000 \text{ W/m}^2$, Find E
 at 0.2 mm of sphere radiation

Q₁ Two charges $q_1 = 3 \mu\text{C}$, $q_2 = -12 \mu\text{C}$
 separated by 2 cm in water
 Find E

Q₂ A charge q accelerated by 30°
 to have speed of light
 Find the charge